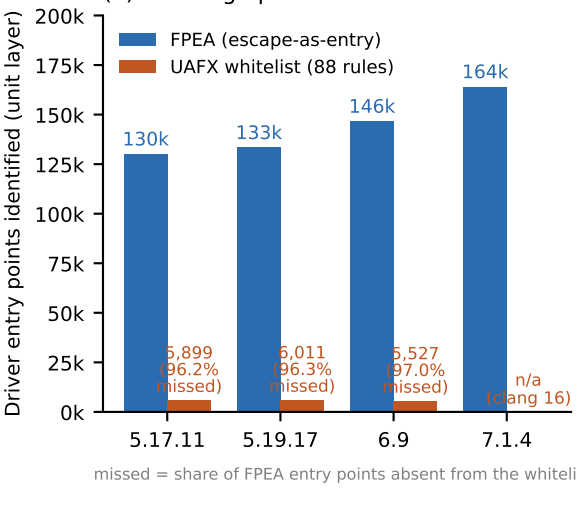
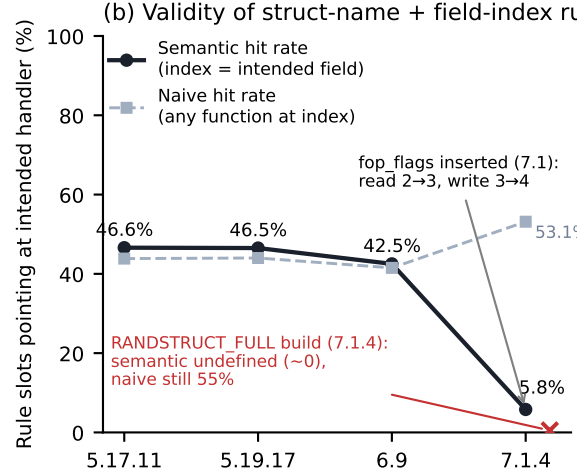

(a) Coverage per kernel version
Driver entry points identified (unit layer)
200k
175k
150k
125k
100k
75k
50k
25k
0k
FPEA (escape-as-entry)
UAFX whitelist (88 rules)
130k
133k
146k
164k
5,899 (96.2% missed)
6,011 (96.3% missed)
5,527 (97.0% missed)
n/a (clang 16)
5.17.11
5.19.17
6.9
7.1.4
missed = share of FPEA entry points absent from the whitelist


(b) Validity of struct-name + field-index rules
Rule slots pointing at intended handler (%)
100
80
60
40
20
0
Semantic hit rate
(index = intended field)
Naive hit rate
(any function at index)
fop_flags inserted (7.1):
read 2→3, write 3→4
46.6%
46.5%
42.5%
53.1%
5.8%
RANDSTRUCT_FULL build (7.1.4):
semantic undefined (~0),
naive still 55%
5.17.11
5.19.17
6.9
7.1.4